Администрирование локальных сетей

Лабораторная работа №4

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Цели и задачи работы

Цель лабораторной работы

Провести подготовительную работу по первоначальной настройке коммутаторов сети и освоить базовые команды конфигурирования сетевого оборудования Cisco.

Ход выполнения работы

Построение топологии сети

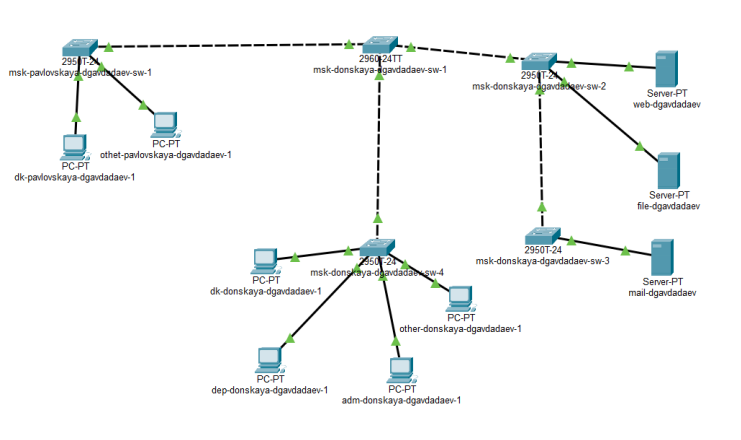

Рис. 1: Топология сети в Packet Tracer

Первоначальная настройка коммутаторов

На каждом коммутаторе была выполнена базовая конфигурация.

Основные этапы настройки: - изменение имени устройства - настройка интерфейса управления VLAN - назначение IP-адреса управления - настройка шлюза по умолчанию - настройка паролей доступа - включение шифрования паролей - создание пользователя администратора - включение удалённого доступа по SSH

Настройка коммутатора msk-donskaya-dgavadadev-sw-1

IOS Command Line Interface

```
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname msk-donskaya-dgavdadaev-sw-1
msk-donskaya-dgavdadaev-sw-1(config)#interface vlan
% Incomplete command.
msk-donskaya-dgavdadaev-sw-1(config)#interface vlan2
msk-donskaya-dgavdadaev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-dgavdadaev-sw-1(config-if)#no shutdown
msk-donskaya-dgavdadaev-sw-1(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-dgavdadaev-sw-1(config-if)#exit
msk-donskaya-dgavdadaev-sw-1(config)#ip default-gateway 10.128.1.1
msk-donskaya-dgavdadaev-sw-1(config)#line vty 0 4
msk-donskaya-dgavdadaev-sw-1(config-line)#password cisco
msk-donskaya-dgavdadaev-sw-1(config-line)#login
msk-donskaya-dgavdadaev-sw-1(config-line)#exit
msk-donskaya-dgavdadaev-sw-1(config)#line console 0
msk-donskaya-dgavdadaev-sw-1(config-line)#password cisco
msk-donskaya-dgavdadaev-sw-1(config-line)#login
msk-donskaya-dgavdadaev-sw-1(config-line)#exit
msk-donskaya-dgavdadaev-sw-1(config)#enable secret cisco
msk-donskaya-dgavdadaev-sw-1(config)#service password-encryption
msk-donskaya-dgavdadaev-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-dgavdadaev-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-dgavdadaev-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dgavdadaev-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dgavdadaev-sw-1(config)#line vty 0 4
*Mar 1 0:10:32.683: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:10:32.683: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dgavdadaev-sw-1(config-line)#transport input ssh
msk-donskaya-dgavdadaev-sw-1(config-line)#exit
msk-donskaya-dgavdadaev-sw-1(config)#exit
msk-donskaya-dgavdadaev-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dgavdadaev-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-dgavdadaev-sw-1#
```

Настройка коммутатора msk-donskaya-dgavadadev-sw-2

IOS Command Line Interface

```
Switch>enable
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hostname msk-donskaya-dgavdadaev-sw-2
msk-donskaya-dgavdadaev-sw-2(config)#
msk-donskaya-dgavdadaev-sw-2(config)#interface vlan2
msk-donskaya-dgavdadaev-sw-2(config-if)#no shutdown
msk-donskaya-dgavdadaev-sw-2(config-if)#ip address 10.128.1.3 255.255.255.0
msk-donskaya-dgavdadaev-sw-2(config-if)#exit
msk-donskaya-dgavdadaev-sw-2(config)#
msk-donskaya-dgavdadaev-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-dgavdadaev-sw-2(config)#
msk-donskaya-dgavdadaev-sw-2(config)#line vty 0 4
msk-donskaya-dgavdadaev-sw-2(config-line)#password cisco
msk-donskaya-dgavdadaev-sw-2(config-line)#login
msk-donskaya-dgavdadaev-sw-2(config-line)#exit
msk-donskaya-dgavdadaev-sw-2(config)#
msk-donskaya-dgavdadaev-sw-2(config)#line console 0
msk-donskaya-dgavdadaev-sw-2(config-line)#password cisco
msk-donskaya-dgavdadaev-sw-2(config-line)#login
msk-donskaya-dgavdadaev-sw-2(config-line)#exit
msk-donskaya-dgavdadaev-sw-2(config)#
msk-donskaya-dgavdadaev-sw-2(config)#enable secret cisco
msk-donskaya-dgavdadaev-sw-2(config)#service password-encryption
msk-donskaya-dgavdadaev-sw-2(config)#username admin privilege 1 secret cisco
msk-donskaya-dgavdadaev-sw-2(config)#
msk-donskaya-dgavdadaev-sw-2(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-dgavdadaev-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dgavdadaev-sw-2.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dgavdadaev-sw-2(config)#
*Mar 1 0:15:22.492: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:15:22.492: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dgavdadaev-sw-2(config)#line vty 0 4
msk-donskaya-dgavdadaev-sw-2(config-line)#transport input ssh
msk-donskaya-dgavdadaev-sw-2(config-line)#
msk-donskaya-dgavdadaev-sw-2(config-line)#exit
msk-donskaya-dgavdadaev-sw-2(config)#exit
```

Настройка коммутатора msk-donskaya-dgavadadev-sw-3

IOS Command Line Interface

```
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hostname msk-donskaya-dgavdadaev-sw-3
msk-donskaya-dgavdadaev-sw-3(config)#
msk-donskaya-dgavdadaev-sw-3(config)#interface vlan2
msk-donskaya-dgavdadaev-sw-3(config-if)#no shutdown
msk-donskaya-dgavdadaev-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
msk-donskaya-dgavdadaev-sw-3(config-if)#exit
msk-donskaya-dgavdadaev-sw-3(config)#
msk-donskaya-dgavdadaev-sw-3(config)#ip default-gateway 10.128.1.1
msk-donskaya-dgavdadaev-sw-3(config)#
msk-donskaya-dgavdadaev-sw-3(config)#line vty 0 4
msk-donskaya-dgavdadaev-sw-3(config-line)#password cisco
msk-donskaya-dgavdadaev-sw-3(config-line)#login
msk-donskaya-dgavdadaev-sw-3(config-line)#exit
msk-donskaya-dgavdadaev-sw-3(config)#
msk-donskaya-dgavdadaev-sw-3(config)#line console 0
msk-donskaya-dgavdadaev-sw-3(config-line)#password cisco
msk-donskaya-dgavdadaev-sw-3(config-line)#login
msk-donskaya-dgavdadaev-sw-3(config-line)#exit
msk-donskaya-dgavdadaev-sw-3(config)#
msk-donskaya-dgavdadaev-sw-3(config)#enable secret cisco
msk-donskaya-dgavdadaev-sw-3(config)#service password-encryption
msk-donskaya-dgavdadaev-sw-3(config)#username admin privilege 1 secret cisco
msk-donskaya-dgavdadaev-sw-3(config)#
msk-donskaya-dgavdadaev-sw-3(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-dgavdadaev-sw-3(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dgavdadaev-sw-3.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dgavdadaev-sw-3(config)#
*Mar 1 0:16:18.736: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:16:18.736: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dgavdadaev-sw-3(config)#line vty 0 4
msk-donskaya-dgavdadaev-sw-3(config-line)#transport input ssh
msk-donskaya-dgavdadaev-sw-3(config-line)#
msk-donskaya-dgavdadaev-sw-3(config-line)#exit
msk-donskaya-dgavdadaev-sw-3(config)#exit
msk-donskaya-dgavdadaev-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-dgavdadaev-sw-3#
%LINK-5-CHANGED: Interface Vlan2, changed state to up
```

Настройка коммутатора msk-donskaya-dgavadadev-sw-4

```
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hostname msk-donskaya-dgavdadaev-sw-4
msk-donskaya-dgavdadaev-sw-4(config)#
msk-donskaya-dgavdadaev-sw-4(config)#interface vlan2
msk-donskaya-dgavdadaev-sw-4(config-if)#no shutdown
msk-donskaya-dgavdadaev-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-donskaya-dgavdadaev-sw-4(config-if)#exit
msk-donskaya-dgavdadaev-sw-4(config)#
msk-donskaya-dgavdadaev-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-dgavdadaev-sw-4(config)#
msk-donskaya-dgavdadaev-sw-4(config)#line vty 0 4
msk-donskaya-dgavdadaev-sw-4(config-line)#password cisco
msk-donskaya-dgavdadaev-sw-4(config-line)#login
msk-donskaya-dgavdadaev-sw-4(config-line)#exit
msk-donskaya-dgavdadaev-sw-4(config)#
msk-donskaya-dgavdadaev-sw-4(config)#line console 0
msk-donskaya-dgavdadaev-sw-4(config-line)#password cisco
msk-donskaya-dgavdadaev-sw-4(config-line)#login
msk-donskaya-dgavdadaev-sw-4(config-line)#exit
msk-donskaya-dgavdadaev-sw-4(config)#
msk-donskaya-dgavdadaev-sw-4(config)#enable secret cisco
msk-donskaya-dgavdadaev-sw-4(config)#service password-encryption
msk-donskaya-dgavdadaev-sw-4(config)#username admin privilege 1 secret cisco
msk-donskaya-dgavdadaev-sw-4(config)#
msk-donskaya-dgavdadaev-sw-4(config)#ip domain-name donskaya.rudn.edu
msk-donskaya-dgavdadaev-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-dgavdadaev-sw-4.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-dgavdadaev-sw-4(config)#
*Mar 1 0:16:57.574: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:16:57.574: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-dgavdadaev-sw-4(config)#line vty 0 4
msk-donskaya-dgavdadaev-sw-4(config-line)#transport input ssh
msk-donskaya-dgavdadaev-sw-4(config-line)#
msk-donskaya-dgavdadaev-sw-4(config-line)#exit
msk-donskaya-dgavdadaev-sw-4(config)#exit
msk-donskaya-dgavdadaev-sw-4#write memory
Building configuration...
[OK]
msk-donskaya-dgavdadaev-sw-4#
%LINK-5-CHANGED: Interface Vlan2, changed state to up
```

Настройка коммутатора msk-pavlovskaya-dgavadadev-sw-1

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/20, changed state to up


Switch>
Switch>enable
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hostname msk-pavlovskaya-dgavdadaev-sw-1
msk-pavlovskaya-dgavdadaev-sw-1(config)#
msk-pavlovskaya-dgavdadaev-sw-1(config)#interface vlan2
msk-pavlovskaya-dgavdadaev-sw-1(config-if)#no shutdown
msk-pavlovskaya-dgavdadaev-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-pavlovskaya-dgavdadaev-sw-1(config-if)#exit
msk-pavlovskaya-dgavdadaev-sw-1(config)#
msk-pavlovskaya-dgavdadaev-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-dgavdadaev-sw-1(config)#
msk-pavlovskaya-dgavdadaev-sw-1(config)#line vty 0 4
msk-pavlovskaya-dgavdadaev-sw-1(config-line)#password cisco
msk-pavlovskaya-dgavdadaev-sw-1(config-line)#login
msk-pavlovskaya-dgavdadaev-sw-1(config-line)#exit
msk-pavlovskaya-dgavdadaev-sw-1(config)#
msk-pavlovskaya-dgavdadaev-sw-1(config)#line console 0
msk-pavlovskaya-dgavdadaev-sw-1(config-line)#password cisco
msk-pavlovskaya-dgavdadaev-sw-1(config-line)#login
msk-pavlovskaya-dgavdadaev-sw-1(config-line)#exit
msk-pavlovskaya-dgavdadaev-sw-1(config)#
msk-pavlovskaya-dgavdadaev-sw-1(config)#enable secret cisco
msk-pavlovskaya-dgavdadaev-sw-1(config)#service password-encryption
msk-pavlovskaya-dgavdadaev-sw-1(config)#username admin privilege 1 secret cisco
msk-pavlovskaya-dgavdadaev-sw-1(config)#
msk-pavlovskaya-dgavdadaev-sw-1(config)#ip domain-name donskaya.rudn.edu
msk-pavlovskaya-dgavdadaev-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-pavlovskaya-dgavdadaev-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-dgavdadaev-sw-1(config)#
*Mar 1 0:17:35.183: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:17:35.183: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-dgavdadaev-sw-1(config)#line vty 0 4
msk-pavlovskaya-dgavdadaev-sw-1(config-line)#transport input ssh
msk-pavlovskaya-dgavdadaev-sw-1(config-line)#
```

Выводы

Итог лабораторной работы

В ходе выполнения лабораторной работы:

- построена топология сети в Cisco Packet Tracer
- выполнена первоначальная настройка всех коммутаторов
- назначены IP-адреса управления
- настроены параметры удалённого доступа
- включено шифрование паролей
- реализован защищённый доступ по SSH